Learning Shape-Proportion Relationships from Labeled Humanoid Cartoons

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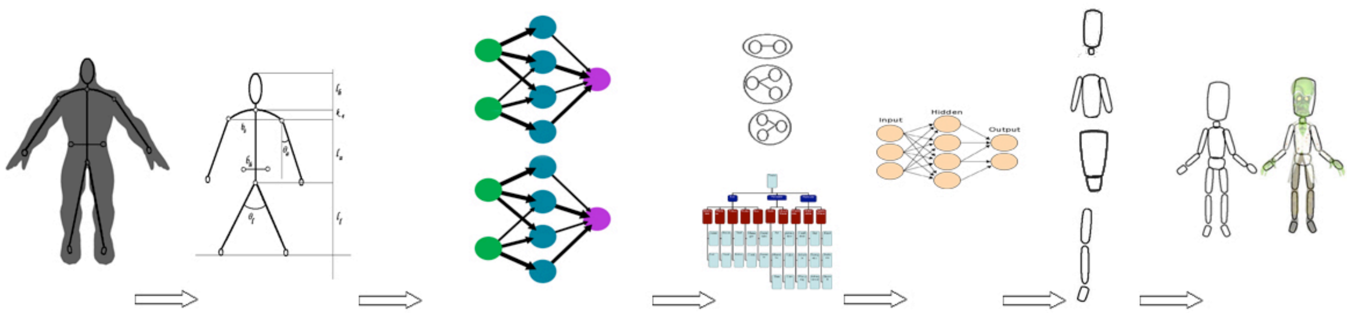

Abstract- **Character design artists typically use shape, pose and proportion as the first design layer to express role, physicality and personality traits. Inspired by this we approach the problem of automatic character synthesis by attempting to learn relations among the body-shape, proportions, pose, and trait labels from finished art. In our prior work [13], we have designed an online game framework to collect and analyze perception data on hundreds of humanoid characters. We clustered the labels and then established a relationship between the body shapes and the pose-proportion feature space. In this paper, we extend the work to explore partial shape synthesis of a character's torso and abdomen, given an input pose and proportion feature set. This paves the way for fully automatic character synthesis from labels. This is an improvement of our prior work, which addressed only shape classification.**

Keywords: Neural networks, Shape-proportion learning, perception modeling, shape synthesis

I. INTRODUCTION

One of the first things children learn is to manipulate and express with primitive shapes. Even as adults, we naturally tend to decompose complex compositions into primitives. Basic shapes like triangles, circles and squares are so well understood, that even a textual/ verbal description of structures in terms of these shapes elicits a natural visualization in our brain. Basic shapes play an important role in design drafts. For example, artists use shape scaffolding to pre-visualize the final form, using basic shapes to represent each component or part. Apart from establishing the volume and mass distribution of the figure, these shapes may also help portray a certain personality, as is widely seen in stylized cartoon drawings. For example, in Pixar's recent animated feature titled UP, the main protagonist had distinctively square features to highlight his cooped-in life. The square features were amplified by contrasting with a large round nose, as well as distinctly rounded supporting characters. Depending on the art style, primitive shapes may become less apparent with the addition of details; e.g. clothes, accessories, and hair for humanoid figures.

We use a vector shape representation that allows consistent blends between square, circle and triangle primitive shapes to express the shapedness of each body-part [12]. We will explain later how this vector primitive based representation of shapes plays a key role in our learning and auto synthesis framework.

The main contribution of this paper is a neural network based expert system that takes as input the body pose and proportions of particular parts and then suggests the vector shape of body parts according to a required character trait. The synthesized shapes will provide a suggestive scaffold for the artists to add details on. By putting the details like clothes, accessories, and hair and drawing the figures in a fuller form the artist gets a humanoid cartoon with a particular character trait. This allows even an amateur or a newbie to create cartoons according to the desired role, physicality, and personality.

We explain the methodology for shape learning and synthesis. Though this can be applied to create independent models for the full body, we demonstrate results for two body parts that tend to show a lot of shape variety, namely the torso and abdomen. We demonstrate the synthesis results for three distinct body types: *fat, thin*, and *muscular*.

II. Literature Review

A variety of shape representation strategies have been used for learning and cognition of visual data. Edelmann and Intrator [5] proposed the use of semantic shape trees to represent well-structured objects like the hammer and airplane. Their goal was to recognize object classes, starting bottom-up with low level features. A drawback of this method is that it needs explicit modeling of the object grammar. Gal et al. [6] propose 2D histogram shape signature that combines two scalar functions defined on the boundary surface, namely a local-diameter and average geodesic distance from one vertex to all other vertices. Though this approach is robust to minor topological changes due to articulation in meshes, the representation lacks intuitiveness.

Classification and regression models on anthropomorphic data have been widely used in the fields of graphics and vision. In most of these models, the feature vectors used are fairly low level; e.g. Cartesian points, curves, distances and moments. Liu et al. [15] perform PCA on low-level point features for original and caricature drawings of human faces. Gooch et al. [9] also cartoonify face photographs by computing Eigenvalues between key facial points after training their system with real face data. Wang et al. [25] used rotational regression to learn deformation offsets of vertices in relation to driver skeletal joints. Meyer and Anderson [17] propose a computation cache for neighborhoods of key points undergoing lighting or deformation calculations, again using PCA analysis on point features. Hsu et al. [11] use CART data mining on body distance measurements (e.g. waist-girth, hip-girth, etc.) and body mass index to classify them into Large/Medium/Small categories for garment production. Marchenko et al. [16] differ from all these approaches by combining ontological metadata (e.g. artist name, style and art period) with low-level image features (e.g. brushwork and color temperature). Though they do not do any shape analysis, they implement a practical learning framework that improves learning results with human-understandable conceptual knowledge layers.

Automatic extraction of information from cartoon images of humanoids poses a number of challenges like perspective distortions, obscured body parts due to posing, and exaggerated non-standard body parts (unlike real humans). We did not find any method that provides a robust solution to this ill-posed problem. Since our goal is not the automation of data collection, which by itself is a significant challenge, we designed a user-friendly system to allow manual annotation of shapes. We derive inspiration from the use of primitive shapes outlined in art books [2,3,10,21] as well as shape perception literature [7,18].

III. Prior Work

In our prior work we gathered human perception on character body shapes through an online game. In this game, as a part of game-play, the players had to tag various cartoon characters with different character traits. The game was designed in such a way that it produced equitable distribution of choice of tags for labeling and the core game-play mechanics ensured that to maximize score the player had to give labels that he think most suitable to the particular cartoon character shown. A set of 140 characters was created by in-house artists, and by annotating images of popular characters over the web. We presented the characters in front-silhouette form so that facial expression, skin tone, dress etc. do not bias the players' judgment. In the silhouette form only the characters pose and body shape–proportions were visible to the players. This way the human perception we collected was directly related to the body shape, pose and proportions.

After this, depending on the trend of trait label assignment to particular character sets we clustered the trait labels we had and got four clusters of labels which had high probability of co-occurring on cartoon characters.
cluster1--->Attractive, Kind
cluster2--->Cute, Shy, Fool, Clerk, Thin
cluster3--->Fat
cluster4--->Muscular, Confident, Superhero

Later we created neural network classifier that could successfully classify characters into one of these four clusters. This classifier accepts 6 input features ($k_{1\text{-}6}$) depicting the given characters pose and body proportions as shown below in Fig 1.

l_h = height of head and neck
l_u = height of upper portion of the body
l_l = height of lower portion of the body
b_h = breadth of abdomen
b_s = width of shoulder
θ_a = angle of arm from the body
θ_l = angle between two legs

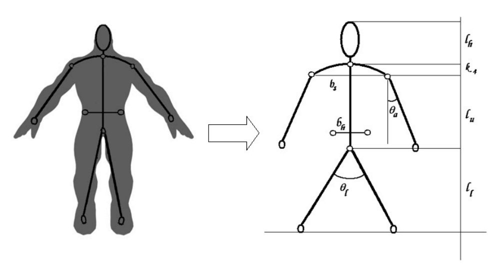

Figure 1: Annotation for pose and proportions [13]

From these features 6 features and ratios are calculated

$k_1 = l_h / (l_u + l_l)$
$k_2 = b_s / b_h$
$k_3 = l_u / l_l$
k_4 = curve(shoulder), calculated from shoulder height
$k_5 = \theta_a$
$k_6 = \theta_l$

In an earlier related work [12], we decomposed the humanoid character into 16 body parts, as shown in Fig. 2. The parts were hand-annotated and then vector fitted to compute the primitive shape weights. Since this representation is compact and intuitive, we showed how it performs better in SVM classification, compared to low level contour features. However, our earlier work was largely dedicated to classification and not synthesis.

Figure 2: Vector annotation of body parts [12]

We take up the shape synthesis problem in this paper. Specifically, we are interested in exploring whether there are relationships between the proportion-pose feature space and the shape of the torso and the abdomen. Using the system-generated shapes as scaffolds, one can draw more impactful finished characters, which will be perceived by most viewers as having the desired character trait(s). We reuse the results of the perception label clustering, and focus on three related body types: *fat, thin,* and *muscular.*

IV. Learning and Synthesis

As described in section III, we have found 4 clusters of traits that co-occur frequently, from human perception data collected in an online game titled *Shadow Shoppe* [13]. We choose the thin, fat and muscular metaphors as they occur in three independent clusters. Each body part in the primitive shape vector has 3 shape weights associated with it namely, square, circle and triangle weights. Since their sum is always 100, we can uniquely describe a shape by specifying any two of these three weight values. In our case we chose square weight and circle weight of given body part.

To predict the square weight of abdomen for fat characters we first train a neural network system on body proportion data that were previously successfully classified as 'fat'. This classifier accepts 6 input features (k_{1-6}) depicting the given characters pose and body proportions (see Section III) and generates an output between 0-100 for the square weight of torso. We had 33 characters in our database, which were automatically classified by our machine as fat. As this is a small number, we used 10-fold cross validation technique (choose ~9x3 samples for training and ~1x3 samples for validation) to train our model. The structure of the network used is shown in Fig 3.

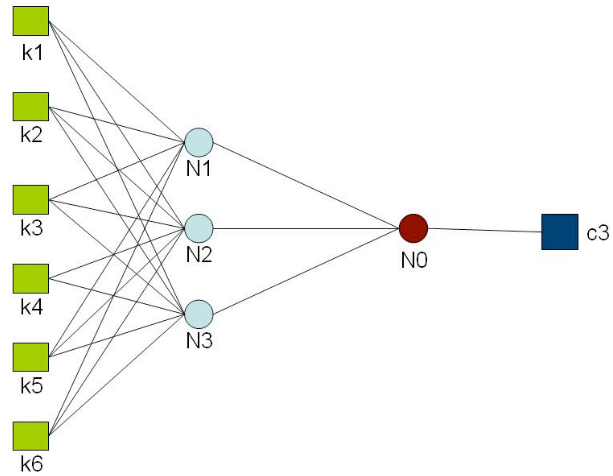

Figure 3: Network Structure for predicting fat torso

The summary of the 10 fold cross validation is:

Correlation coefficient	0.8696
Mean absolute error	8.8701
Root mean squared error	11.2042
Total Number of Instances	33

A mean absolute error of 8.8 is acceptable in our case because a change of 8 in shape weight does not create any drastic change in shape perception of the overall shape.

The Training parameters were:

Number of sigmoid nodes	3
Learning rate	0.2
Momentum	0.2

Various node-link weights and threshold values are given in the Tables 1 and 2 below:

Table 1: Link weights and thresholds for sigmoid nodes

Sigmoid Node	Threshold	k1	k2	k3	k4	k5	k6
Node 1	-6.23	-3.28	11.83	8.97	-1.53	-1.79	0.38
Node 2	6.623	-1.16	-3.14	-3.61	-3.78	9.41	-3.35
Node 3	0.67	-3.18	-2.14	-2.63	-2.99	6.12	-4.67

Table 2: Link weights and threshold for linear node

Linear Node	Threshold	Node 1	Node 2	Node 3
Node 0	2.001	-7.743	-3.035	1.325

We similarly train another model for predicting the circle weight for fat torso, with the summary of accuracy as follows:

Correlation coefficient	0.953
Mean absolute error	6.2164
Root mean squared error	8.0947
Total Number of Instances	33

Training parameters used:

Number of sigmoid node	3
Learning rate	0.3
Momentum	0.2

As illustrated in Fig 4, a combined application of these two neural networks on body proportion and pose data (k_{1-6} values) creates a shape vector for 'fat torso'.

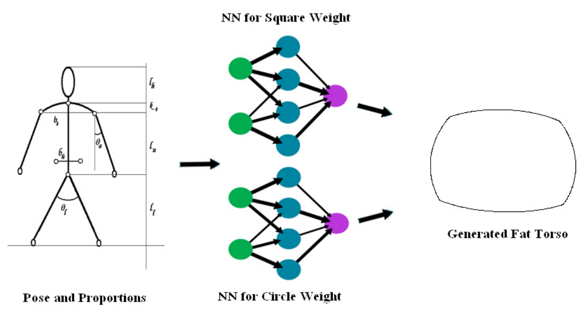

Figure 4: Synthesis of "fat" torso shape

Similarly we train two neural networks to generate abdomen for fat characters. Since the structure and methodology of these networks are quite similar to ones for torso described above, we omit the details here. Readers interested in implementing the synthesis models can download the detailed Neural Network structure and weight information for torso and abdomen shapes for all three body types, from [28]. Tables 4 and 5 summarize the accuracy of the shape classifiers for thin and muscular and thin body types respectively.

Table 4: Accuracy for Shape Classifiers for 26 Muscular Test samples

Muscular Torso Square Weight	Correlation coefficient	0.83
	Mean absolute error	6.71
	Root mean squared error	8.48
	Relative absolute error	54.89 %
	Root relative squared error	56.94%
Muscular Torso Circle Weight	Correlation coefficient	0.72
	Mean absolute error	11.23
	Root mean squared error	15.10
	Relative absolute error	73.38 %
	Root relative squared error	80.37%
Muscular Abdomen Square Weight	Correlation coefficient	0.97
	Mean absolute error	3.99
	Root mean squared error	5.07
	Relative absolute error	26.71 %
	Root relative squared error	29.18 %
Muscular Abdomen Circle Weight	Correlation coefficient	0.82
	Mean absolute error	10.16
	Root mean squared error	14.08
	Relative absolute error	52.52 %
	Root relative squared error	59.28 %

Table 5: Accuracy for Shape Classifiers for 34 Thin Test samples

Thin Torso Square Weight	Correlation coefficient	0.60
	Mean absolute error	15.41
	Root mean squared error	19.01
	Relative absolute error	90.93%
	Root relative squared error	91.41%
Thin Torso Circle Weight	Correlation coefficient	0.74
	Mean absolute error	11.98
	Root mean squared error	14.61
	Relative absolute error	80.15%
	Root relative squared error	79.74%
Thin Abdomen Square Weight	Correlation coefficient	0.96
	Mean absolute error	3.18
	Root mean squared error	5.68
	Relative absolute error	23.89 %
	Root relative squared error	28.95 %
Thin Abdomen Circle Weight	Correlation coefficient	0.91
	Mean absolute error	7.16
	Root mean squared error	10.09
	Relative absolute error	41.34 %
	Root relative squared error	43.59 %

As can be seen from the above tables, the Neural Networks were able to fairly accurately mimic the shapes from the small training set. These models can now synthesize a smooth range of shape values, given subtle changes in the input pose/shape vector. We can now use interesting techniques like Genetic Evolution and stochastic noise to create a variety of character shapes from a family of characters.

Fig. 5 shows the synthesized torso shapes for the three body types, as blue overlapped cages over the test characters. The ground truth here is the fitted manual annotation data. As can be seen, our machine is able to synthesize shapes that visually resemble the ground truth quite closely.

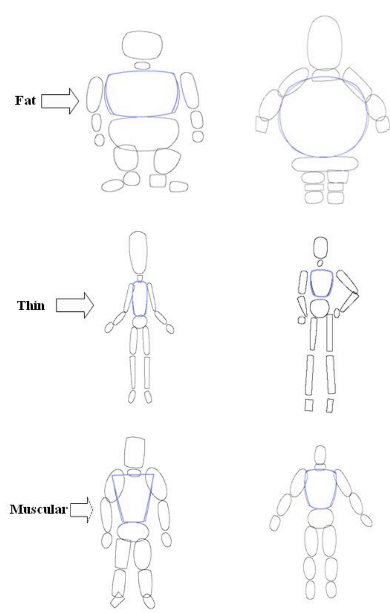

Figure 5: Visual Comparison of Shape Synthesis

V. Conclusion

We have extended our earlier work in automatically clustering character labels and figure associations to these cluster labels, from just classification of new body shapes, towards shape synthesis from text labels. We have illustrated a simple Neural Network framework that can learn shape synthesis for different character body types. We have achieved decent results from a relatively small data set, and are able to test it on other smaller independent drawing sets. We have provided the model details of our torso and abdomen synthesizers online, for readers to download and try out our synthesis model. We hope to train full body synthesis models soon. One of the limitations of our current approach is that we are synthesizing body shapes independently from each other. We might need to model additional neighborhood relationship constraints (in the spirit of ideas in Gistalt theory [18]) for consistent generation of full-bodied characters.

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